

ER Physiology Practice Questions with answers

Topic: Adrenal gland

1. A 39-year-old woman comes to the office because of an 8-month history of red streaks over her chest and abdomen and a 5-kg (11-lb) weight gain. She has no edema or shortness of breath. Her blood pressure is 144/92 mm Hg; other vital signs are within normal limits. Physical examination shows facial puffiness, truncal obesity, violaceous striae over abdomen and breasts, thin extremities, and increased abdominal girth. Laboratory studies show increased serum cortisol concentration. Skeletal x-ray shows osteoporosis. Which of the following additional laboratory findings is most likely in this patient?

- A. Hyperkalemia
- B. Hypercalcemia
- C. Hyperglycemia
- D. Hyponatremia
- E. Elevates serum bilirubin

2. A 38-year-old woman comes to the physician because of a 5-month history of fatigue, muscle weakness, and recurrent episodes of muscle cramps and headaches. Her blood pressure is 168/96 mm Hg; other vital signs are within normal limits. Physical examination shows no abnormalities. Serum studies show elevated concentrations of sodium, bicarbonate, and aldosterone but reduced potassium. MRI of the abdomen shows a left adrenal mass. Which of the following renal mechanisms best explains a low serum potassium in this patient?

- A. Increased glomerular filtration
- B. Reduced potassium excretion
- C. Reduced intercalated cell activity in the collecting tubules
- D. Increased ENaC activity in the principal cells
- E. Increased Na-Cl symporter activity in the distal convoluted tubules

3. A 43-year-old woman comes to the physician because of a 6-month history of episodes of palpitations, anxiety, and sweating. Her family history is significant for multiple endocrine neoplasia (MEN) syndrome. Her blood pressure is 155/94 mmHg. Laboratory studies show elevated serum and urinary concentrations of vanillylmandelic acid. Which of the following cell hypersecretion is most likely in this patient?

- A. Zona glomerulosa cells
- B. Zona fasciculata cells
- C. Zona reticularis cells
- D. Chromaffin cells
- E. Corticotrope cells

4. A 35-year-old woman comes to the physician because of a 5-month history of weight gain, muscle weakness, and bone pains. She has a 16-year history of bronchial asthma and started using prednisolone (synthetic glucocorticoid) inhaler for the last 6 months. Her blood pressure is 155/95 mm Hg; other vital signs are within normal limits. Physical examination shows round face, truncal obesity with reddish purple striae over the abdomen, relatively thinner arms and legs, and unhealed wound on her left leg. Laboratory studies show an abnormal glucose tolerance test and an extremely low concentration of adrenocorticotrophic hormone (ACTH). Skeletal radiography shows osteoporosis. She is diagnosed to have Cushing syndrome because of the steroid inhaler and advised to stop it. Which of the following mechanisms best explains reduced serum ACTH in this patient?

- A. Stimulation of the adrenal cortex by exogenous steroid
- B. Suppression of the hypothalamic-pituitary-adrenal axis
- C. Negative feedback suppression of the hypothalamus by excess cortisol
- D. Negative feedback suppression of the anterior pituitary by excess cortisol
- E. Stimulation of the liver to increase steroid-binding globulin

5. A 53-year-old man comes to the physician because of a 1-week history of generalized weakness, nausea, loss of appetite, dizziness, and a low urine output. He also has a 12-month history of a 10-kg (22-lb) weight loss, intermittent abdominal pain, and recurrent infections. Five years ago, he was diagnosed with HIV positive, and he is not adherent to treatment. His pulse is 88/min and blood pressure is 90/60 mm Hg. Laboratory studies show plasma sodium of 122 mmol/L (Normal: 135–145 mmol/L), potassium of 6.9 mmol/L (Normal: 3.5–5.2 mmol/L), low serum glucose, low serum bicarbonate, and an extremely low serum aldosterone concentration. Which of the following sets of laboratory alterations and clinical features is most likely in this patient?

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B	↓	↑	Metabolic alkalosis	Truncal obesity
C	↓	↓	Respiratory acidosis	Hyperpigmentation
D	↑	↓	Respiratory alkalosis	Truncal obesity
E	↓	↑	Mixed acidosis	Hyperpigmentation
F	↑	↓	Mixed alkalosis	Hypopigmentation

- A. A
- B. B
- C. C
- D. D
- E. E
- F. F

6. A 45-year-old woman comes to the physician because of a 4-day history of chest pain, fatigue, and peripheral numbness. Her blood pressure is 162/94 mm Hg. Physical examination shows no abnormalities. An ultrasound of her kidneys shows abnormal enlargement of her left adrenal gland. Laboratory studies of an adrenal biopsy sample indicate overexpression of aldosterone synthase. Which of the following regions of the adrenal gland would most likely express this enzyme?

- A. Zona glomerulosa
- B. Zona fasciculata
- C. Zona reticularis
- D. Chromaffin cells
- E. Corticotrope cells

7. A 24-year-old man comes to the physician because of a 3-week history of fatigue, headache, and body pain. Physical examination shows hypertension, moon face, thin peripheral extremities, and abdominal fat deposition. Laboratory studies show elevated plasma ACTH level. Which of the following second messenger systems does ACTH activate after binding to its receptor in the cells of the adrenal cortex?

- A. IP3 and DAG
- B. cAMP
- C. Tyrosine kinase
- D. Steroid acute regulatory protein (STAR)
- E. cGMP

8. A 31-year-old woman comes to the physician because of a 4-week history of fatigue, frequent headaches, and a 7-kg (15.4-lb) weight gain. Her pulse is 86/min and blood pressure is 158/96 mm Hg. Physical examination shows moon face, thin peripheral extremities, and abdominal fat deposition. Laboratory studies show elevated plasma cortisol, elevated urinary cortisol, a very low plasma ACTH, and normal serum sodium and potassium. Which of the following is the most likely diagnosis?

- A. Cushing disease
- B. Conn syndrome
- C. Pheochromocytoma
- D. Cushing syndrome
- E. Addison disease

9. A 25-year-old man comes to the physician because of a 2-week history of weakness and fatigue. He also has frequent feelings of numbness in his limbs. His blood pressure is 158/95 mmHg, and pulse is 84/min. Laboratory studies show an elevated serum renin. Renal ultrasonography shows a mass in the left kidney. Histological examination

of renal biopsy shows uncontrolled growth of the juxtaglomerular cells. Which of the following additional hormones is most likely elevated in this patient?

- A. Androstenedione
- B. Aldosterone
- C. Cortisol
- D. Epinephrine
- E. Norepinephrine

10. A 28-year-old man comes to the physician because of a 2-day history of anxiety and nervousness. He was diagnosed with panic disorder 1 year ago. His blood pressure is 158/90 mmHg, and pulse is 102/min. Laboratory studies show an elevated plasma cortisol level. He is found to have a stress-induced stimulation of adrenal gland. Which of the following physiological effects is most likely in this patient?

- A. Decreased blood glucose
- B. Suppression of the immune system
- C. Pupillary constriction
- D. Bronchoconstriction
- E. Release of norepinephrine

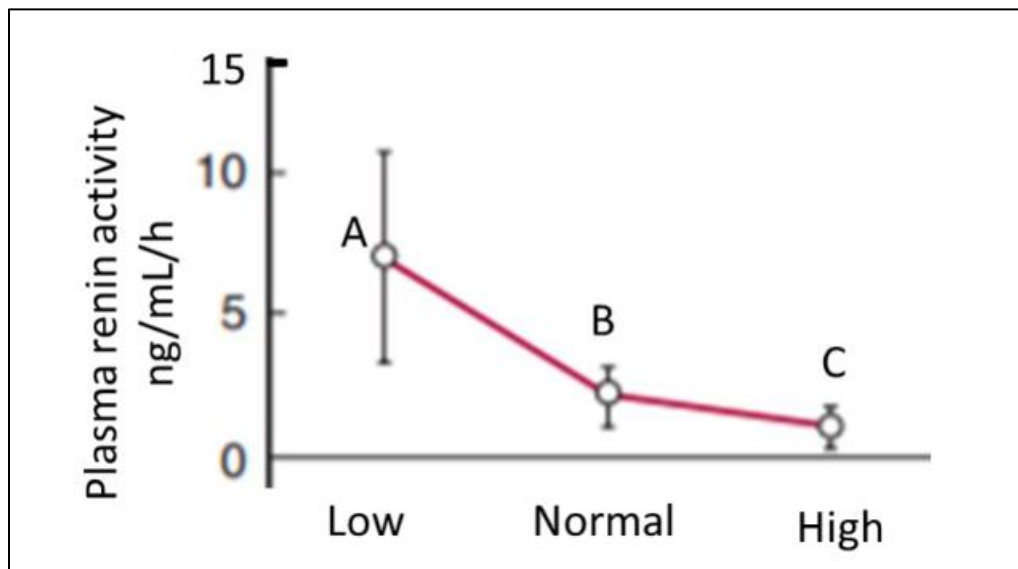
11. A 44-year-old man comes to the physician for a follow up examination. He was diagnosed with Addison disease 6 months ago. Which of the following findings is most likely in this patient?

- A. Hyperpigmentation of the skin
- B. Increased serum cortisol level
- C. Increased serum adrenal androgen level
- D. Decreased serum ACTH level
- E. Hypertension

12. A 36-year-old woman comes to the physician because of a 4-week history of frequent headaches and shortness of breath on exertion. Her pulse is 84/min and blood pressure is 164/96 mm Hg. Physical examination shows loud second heart sound. Laboratory studies show elevated serum aldosterone and low serum renin. She is diagnosed with primary aldosteronism. A decrease in which of the following is most likely in this patient's blood?

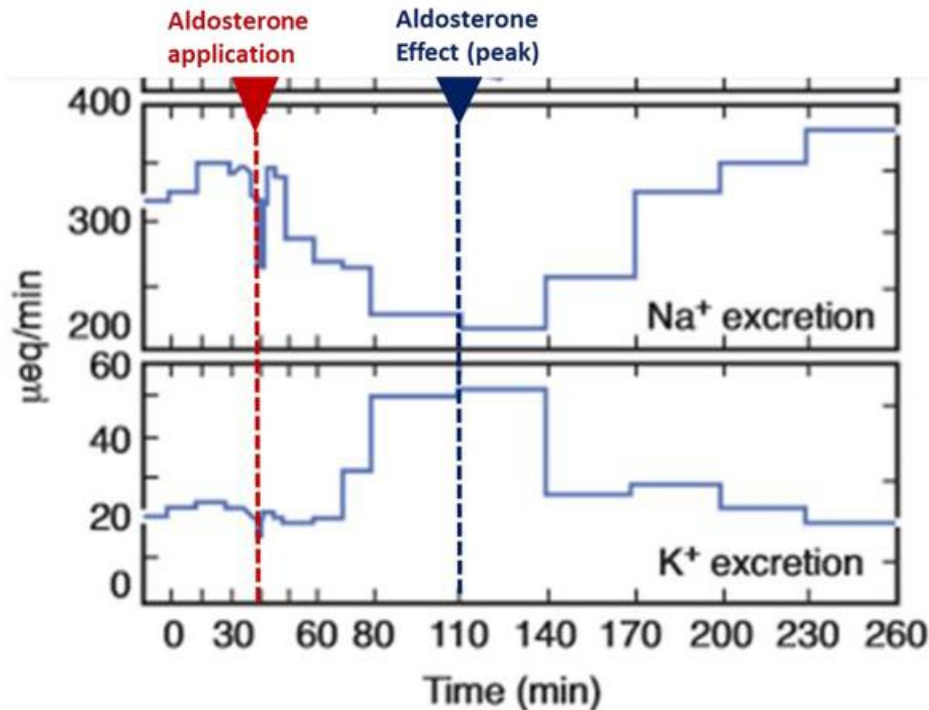
- A. Sodium
- B. Calcium
- C. Potassium
- D. Bicarbonate
- E. Arterial pH

13. A study is conducted on a cohort of healthy volunteers to find out the effect of low-, normal-, and high-sodium intake in diet on plasma renin activity. The figure below shows the result of this study. Point A, B, and C represent the plasma renin activity at low-, normal-, and high-sodium diet respectively. Which of the following additional physiological effects is most likely in response to a high-sodium diet?



- A. Plasma angiotensin II increases
- B. Plasma aldosterone decreases
- C. Urinary sodium excretion decreases
- D. Urinary potassium excretion increases
- E. Plasma ADH decreases

14. A study is conducted to target the molecular mechanism of action of aldosterone in the renal tubules. The figure below shows the time dependent urinary excretion of Na^+ and K^+ in response to the direct administration of a test dose of aldosterone into the aorta of a subject. It has been noticed from the figure that there is a time lag between the application of aldosterone and onset of its peak effect. Which of the following best explains this time lag?



- A. Synthesis of new proteins by genomic mechanism
- B. Transport of Na⁺ from DCT lumen to plasma
- C. Transport of K⁺ plasma to DCT lumen
- D. Action of Angiotensin II on the adrenal cortex
- E. Action of ACTH on the adrenal cortex

15. A 35-year-old woman comes to the physician for a routine examination. She consumes a diet high in fat, sodium, and sugar. Her pulse is 84/min and blood pressure is 168/96 mm Hg. Physical examination shows a loud second heart sound and an epigastric bruit. Laboratory studies show elevated serum total cholesterol and LDL-cholesterol. Imaging studies show a stenosis of the right renal artery. Increased secretion of which of the following hormones is the most likely cause of hypertension in this patient?

- A. Cortisol
- B. Aldosterone
- C. Androgens
- D. Catecholamines
- E. ADH

ANSWER KEY AND EXPLANATION

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- A. Hyperkalemia
- B. Hypercalcemia
- C. Hyperglycemia
- D. Hyponatremia
- E. Elevates serum bilirubin

Answer: C. Hyperglycemia. This patient has Cushing syndrome due to elevated cortisol. Increase in serum glucose is one of the laboratory findings in these patients. This is because cortisol increases gluconeogenesis and glycogenolysis in the liver and decreases glucose utilization by the peripheral tissues.

2. A 38-year-old woman comes to the physician because of a 5-month history of fatigue, muscle weakness, and recurrent episodes of muscle cramps and headaches. Her blood pressure is 168/96 mm Hg; other vital signs are within normal limits. Physical examination shows no abnormalities. Serum studies show elevated concentrations of sodium, bicarbonate, and aldosterone but reduced potassium. MRI of the abdomen shows a left adrenal mass. Which of the following renal mechanisms best explains a low serum potassium in this patient?

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Answer: D. Increased ENaC activity in the principal cells. This patient has primary hyperaldosteronism because of a tumor that secretes aldosterone. Aldosterone increases the activity of the ENaC (epithelial sodium channels) in the principal cells of the distal nephron. This results in increased sodium reabsorption (hence, elevated serum sodium) and increased potassium loss in urine (hence reduced serum potassium). Aldosterone also increases H⁺-ATPase activity in the intercalated cells that increases acid loss and metabolic alkalosis (hence elevated serum bicarbonate).

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Answer: D. Chromaffin cells. This patient has pheochromocytoma which is the tumor of the adrenal medulla. The chromaffin cells hypersecrete catecholamines that causes hypertension, nervousness, palpitations, and sweating.

4. A 35-year-old woman comes to the physician because of a 5-month history of weight gain, muscle weakness, and bone pains. She has a 16-year history of bronchial asthma and started using prednisolone (synthetic glucocorticoid) inhaler for the last 6 months. Her blood pressure is 155/95 mm Hg; other vital signs are within normal limits. Physical examination shows round face, truncal obesity with reddish purple striae over the abdomen, relatively thinner arms and legs, and unhealed wound on her left leg. Laboratory studies show an abnormal glucose tolerance test and an extremely low concentration of adrenocorticotrophic hormone (ACTH). Skeletal radiography shows osteoporosis. She is diagnosed to have Cushing syndrome because of the steroid inhaler and advised to stop it. Which of the following mechanisms best explains reduced serum ACTH in this patient?

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- E. Stimulation of the liver to increase steroid-binding globulin

Answer: B. Suppression of the hypothalamic-pituitary-adrenal axis. This patient takes synthetic glucocorticoid that functions like the naturally secreted cortisol from the adrenal cortex. All clinical features of this patient are same as that occurs with excess cortisol secretion. Like cortisol, prednisolone also inhibits hypothalamus and anterior pituitary. This causes suppression of the hypothalamic-pituitary-adrenal (HPA) axis. Long term use of such exogenous steroid reduces the size (atrophy) of the cortisol-secreting zones of the adrenal cortex.

5. A 53-year-old man comes to the physician because of a 1-week history of generalized weakness, nausea, loss of appetite, dizziness, and a low urine output. He also has a 12-month history of a 10-kg (22-lb) weight loss, intermittent abdominal pain, and recurrent infections. Five years ago, he was diagnosed with HIV positive, and he is not adherent to treatment. His pulse is 88/min and blood pressure is 90/60 mm Hg. Laboratory studies show plasma sodium of 122 mmol/L (Normal: 135–145 mmol/L), potassium of 6.9 mmol/L (Normal: 3.5–5.2 mmol/L), low serum glucose, low serum bicarbonate, and an extremely low serum aldosterone concentration. Which of the following sets of laboratory alterations and clinical features is most likely in this patient?

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- D. D
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Answer: Option A. This patient has Addison disease (primary adrenocortical deficiency) because of untreated HIV disease. He has deficiency of both cortisol and aldosterone. Low cortisol increases serum ACTH concentration which produces hyperpigmentation of the skin and mucous membranes. Low aldosterone and low BP stimulates renin secretion (serum renin increases). Low aldosterone causes low serum sodium a high serum potassium (hyperkalemia). Hyperkalemia and reduced urinary acid loss produces metabolic acidosis (low serum bicarbonate).

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Answer: A. Zona glomerulosa layer of the adrenal cortex alone expresses aldosterone synthase enzyme.

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- A. IP3 and DAG
- B. cAMP
- C. Tyrosine kinase
- D. Steroid acute regulatory protein (STAR)
- E. cGMP

Answer: B. ACTH is a peptide hormone and binds to a receptor on the cell membrane and activates Gs protein-adenylyl cyclase and stimulates production of cAMP as the second messenger.

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- A. Cushing disease
- B. Conn syndrome
- C. Pheochromocytoma
- D. Cushing syndrome
- E. Addison disease

Answer: D. This patient has features of hypercortisolism, a clinical condition known as Cushing syndrome. Cushing disease is the clinical condition in which hypersecretion of ACTH from the anterior pituitary produces Cushing syndrome. In this patient, ACTH is very low which indicates that the source of cortisol excess is from the cortisol-secreting tumor of the adrenal cortex.

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and fatigue. He also has frequent feelings of numbness in his limbs. His blood pressure is 158/95 mmHg, and pulse is 84/min. Laboratory studies show an elevated serum renin. Renal ultrasonography shows a mass in the left kidney. Histological examination of renal biopsy shows uncontrolled growth of the juxtaglomerular cells. Which of the following additional hormones is most likely elevated in this patient?

- A. Androstenedione
- B. Aldosterone
- C. Cortisol
- D. Epinephrine
- E. Norepinephrine

Answer: B. This patient has a renin secreting tumor in the kidney. Renin activates the renin-angiotensin-aldosterone system resulting in increased production of aldosterone.

10. A 28-year-old man comes to the physician because of a 2-day history of anxiety and nervousness. He was diagnosed with panic disorder 1 year ago. His blood pressure is 158/90 mmHg, and pulse is 102/min. Laboratory studies show an elevated plasma cortisol level. He is found to have a stress-induced stimulation of adrenal gland. Which of the following physiological effects is most likely in this patient?

- A. Decreased blood glucose
- B. Suppression of the immune system
- C. Pupillary constriction
- D. Bronchoconstriction
- E. Release of norepinephrine

Answer: B. Stress stimulates both sympathetic system and cortisol secretion. Elevated cortisol produces immune suppression. Other physiologic effects during stress include an increase in heart rate, blood pressure, blood glucose, pupillary dilation, and bronchodilation.

11. A 44-year-old man comes to the physician for a follow up examination. He was diagnosed with Addison disease 6 months ago. Which of the following findings is most likely in this patient?

- A. Hyperpigmentation of the skin
- B. Increased serum cortisol level
- C. Increased serum adrenal androgen level
- D. Decreased serum ACTH level
- E. Hypertension

Answer: A. Addison disease is associated with atrophy of the adrenal gland and hence decreased secretion of cortisol, adrenal androgens, and aldosterone. Low cortisol

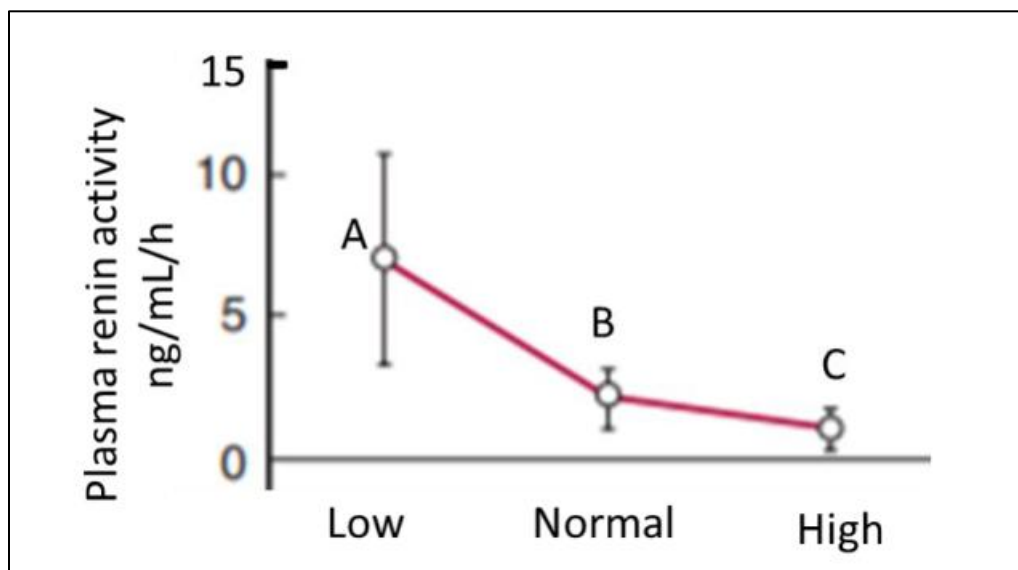
increases serum ACTH level (withdrawal of the negative feedback). Elevated ACTH causes increased melanin synthesis in the skin and hyperpigmentation.

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- A. Sodium
- B. Calcium
- C. Potassium
- D. Bicarbonate
- E. Arterial pH

Answer: C. High aldosterone causes retention of sodium and loss of potassium in urine. A low potassium produces metabolic alkalosis where arterial pH and serum bicarbonate are high.

13. A study is conducted on a cohort of healthy volunteers to find out the effect of low-, normal-, and high-sodium intake in diet on plasma renin activity. The figure below shows the result of this study. Point A, B, and C represent the plasma renin activity at low-, normal-, and high-sodium diet respectively. Which of the following additional physiological effects is most likely in response to a high-sodium diet?

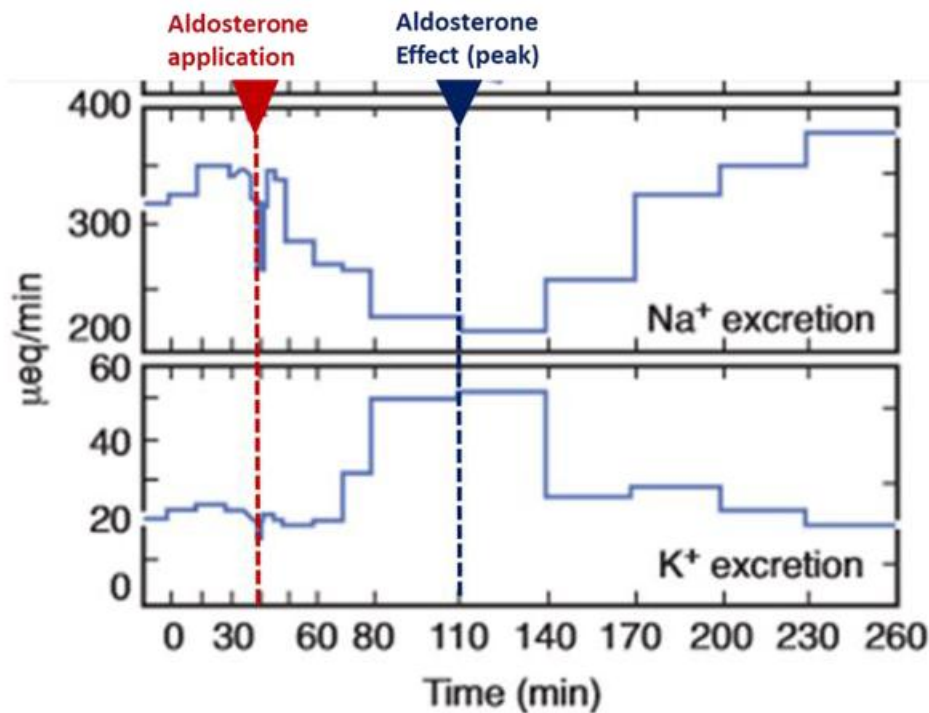


- A. Plasma angiotensin II increases
- B. Plasma aldosterone decreases

- C. Urinary sodium excretion decreases
- D. Urinary potassium excretion increases
- E. Plasma ADH decreases

Answer: B. A high-sodium diet (point C in the figure) decreases plasma renin activity. High dietary sodium expands ECF volume by water reabsorption in GIT and vasopressin mechanism (because of change in plasma osmolality). Increased ECF (blood) volume decreases the renin release from the kidney as there is increased renal perfusion. Decreased renin decreases the angiotensin II formation and thus decreases the aldosterone secretion from the adrenal cortex. A low plasma aldosterone favors renal excretion of sodium and decreases renal excretion of potassium.

14. A study is conducted to target the molecular mechanism of action of aldosterone in the renal tubules. The figure below shows the time dependent urinary excretion of Na^+ and K^+ in response to the direct administration of a test dose of aldosterone into the aorta of a subject. It has been noticed from the figure that there is a time lag between the application of aldosterone and onset of its peak effect. Which of the following best explains this time lag?



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- B. Transport of Na^+ from DCT lumen to plasma
- C. Transport of K^+ plasma to DCT lumen
- D. Action of Angiotensin II on the adrenal cortex
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Answer: A. Aldosterone binds to the cytoplasmic receptors in the principal cells of the

distal nephron. The aldosterone-receptor complex moves to the nucleus and stimulates transcription of mRNA, and the synthesis of a new protein called epithelial sodium channel (ENaC). These proteins are incorporated into the luminal side of the principal cells and increase sodium reabsorption. The entire process takes a certain time and therefore there is a time lag.

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- A. Cortisol
- B. Aldosterone
- C. Androgens
- D. Catecholamines
- E. ADH

Answer: B. This patient has secondary hyperaldosteronism due to stenosis of the renal artery. The major consequence of this stenosis is an increased renin secretion from kidney (JG cells) leading to the increased secretion of aldosterone from adrenal cortex.